

Stable Manifold Partition Algorithm

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The goal here is to take the stable manifolds and create two partitions based on the holes we punch between pseudoneighbor pairs. One partition will correspond to the right side of the stable manifold, and the other will correspond to the left. Here, we set the orientation as the dynamical direction of the manifold. For the stable manifold the orientation is set looking towards the anchor point.

To begin partitioning, we start by punching holes based on the computed pseudoneighbor pairs. For each pair, we punch a hole in the region bounded by the stable and unstable manifolds. The left partition will be determined based on the holes punched on the left side of the stable manifold, and the right partition will be determined based on the holes punched on the right side of the stable manifold.

To generate all the necessary holes and not compute redundant holes, we take the initial pseudoneighbors we found with our algorithm and refer to them as reference pseudoneighbors. This is the set of pseudoneighbors found at the most extreme end of the stable manifold. Then, each reference pseudoneighbor pair is associated with the unstable "reference" bridge they are connected to. Next, each reference bridge is mapped backward to determine which bridge it maps into. Repeat this process recursively until the bridge begins to map onto itself. At that point, no new topologically significant holes will be generated from that reference pseudoneighbor trajectory. This must be done with each reference pseudoneighbor in the entire computed trellis. Once that is complete, all topologically significant holes will have been punched in the phase space.

Now we begin to partition the stable manifold. Take each bridge containing a pseudoneighbor and determine the endpoints of those bridges. We define two sets L and R that contain all such endpoints for pseudoneighbors that generated a hole on the left or right side of the stable manifold, respectively. At this point, the partitioning will continue identically for both sets, so we will focus only on the right partition based on the intersections contained within R . Know that the left partition is created the same way from the intersections contained within L . Now, begin at the anchor point

We will start with the active partition. Take each hole in the trellis within the resonance zone. Find the two intersections bounding the region the hole is within. Do this for each hole in the trellis until you have a set of "interior" intersections. Then for beginning at the anchor point start to define the partition intervals. At each intersection define the boundary of a new interval. The intervals which bound a hole should be open intervals (ie they do not contain the intersection points).

fundamental algorithm to take a bridge map it backwards and figure out which bridge it lies within

* take the reference inert bridges with the pseudoneighbors and map them backwards until they continue to map within the same bridge, then you have generated all the holes necessary *

* think about the singleton bridges with a hole on either side *

Algorithm 1 PARTITION-STABLE-MANIFOLD(T) - Returns two partitions of the stable manifold

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1: Let  $P$  be the set of all pseudoneighbor pairs in the trellis
2: Find the intersections  $N$  which define the bridges the pseudoneighbor pairs lies on
3: Punch holes within the regions bounded by the stable and unstable manifolds attached to  $N$ 
4: Split the intersections into two groups. Intersections which define an exterior hole  $N_E$  and intersection
   which define an interior hole  $N_I$ . Note these should also include the anchor point.
5: initial_boundary = 0
6: for each point in  $N_I$  do
7:   if  $N_{I_i}$  and  $N_{I_{(i+1)}}$  bound an interior hole then
8:     stable_partition =  $(N_{I_i}, N_{I_{(i+1)}})$ 
9:   else
10:    stable_partition =  $[N_{I_i}, N_{I_{(i+1)}}]$ 
11:   end if
12: end for

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